

Emerging Technologies -- Guided Notes

Unit tp-4.7 | CompTIA Network+ N10-009 Obj. FC0-U71 4.7 | N10-008 FC0-U71 4.7

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

Augmented Reality (AR) _____ digital content onto the real world (visible through a phone camera or AR glasses). Virtual Reality (VR) creates a fully _____ digital environment that replaces the physical world. Mixed Reality (MR) allows digital objects to interact with _____ world objects.

Question 2

Blockchain is a _____-distributed ledger that records transactions across multiple nodes. Each block contains a _____ of the previous block, creating a chain that makes tampering _____ to detect. It is the underlying technology for _____ currencies like Bitcoin and Ethereum.

Question 3

Quantum computing uses _____ (quantum bits) that can exist as 0, 1, or both simultaneously (superposition). This allows quantum computers to evaluate _____ exponentially more solutions at once than classical computers. A sufficiently powerful quantum computer could break current _____ encryption (RSA/ECC), requiring _____-resistant algorithms.

Question 4

5G is the 5th generation cellular standard. Compared to 4G LTE, 5G offers _____ speeds, _____ latency, and supports far more _____ devices simultaneously. Use cases include smart city infrastructure, autonomous vehicles, and _____ (remote robotic surgery). 5G mmWave offers highest speeds but limited _____ penetration.

Question 5

Edge computing moves processing closer to where data is _____, reducing _____ and bandwidth usage. It complements rather than replaces cloud computing. Industrial IoT sensors, self-driving cars, and smart factories use edge processing because they cannot tolerate the _____ ms round-trip delay to a remote cloud server for real-time decisions.

Question 6

3D printing (additive manufacturing) builds _____ objects layer by layer from a digital _____ (STL file). It enables rapid _____ (quick physical models of designs), customized medical implants, and on-demand part replacement. The two main materials are _____ (plastic filament, FDM) and _____ (photopolymer, resin).

Question 7

Automation replaces repetitive human tasks with machines or software. _____ Process Automation (RPA) uses software robots to automate rule-based digital tasks like data entry. _____ automation uses physical robots for repetitive factory tasks. The societal concern is _____ displacement -- jobs replaced without equivalent new jobs being created.

Question 8 -- Real World Application

REAL WORLD: A small manufacturing company is considering adopting robotics to automate their assembly line. Currently 12 workers perform repetitive welding and packaging tasks. A manager argues the robots will pay for themselves in 3 years and improve product consistency. A workers representative argues it will harm the employees. Describe two legitimate benefits and two legitimate concerns, then suggest one way the company could balance both interests.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/tp-4-7-emerging-technologies/  
tp-4-7-emerging-technologies-answer-key.pdf
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